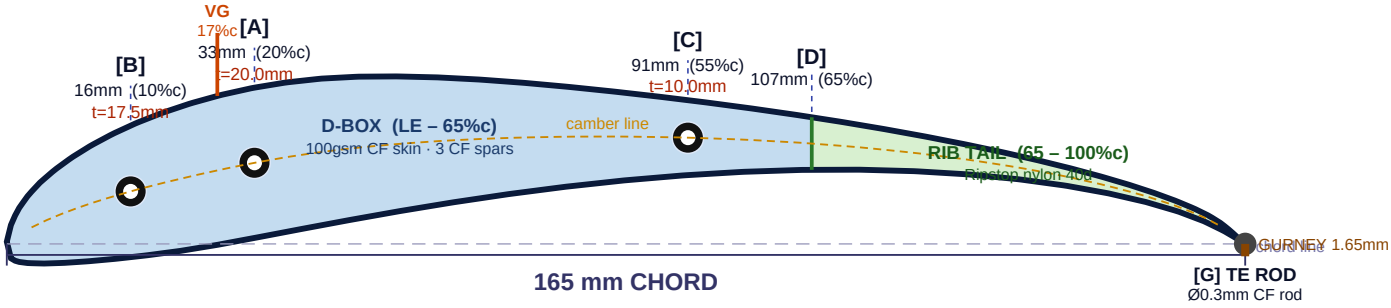
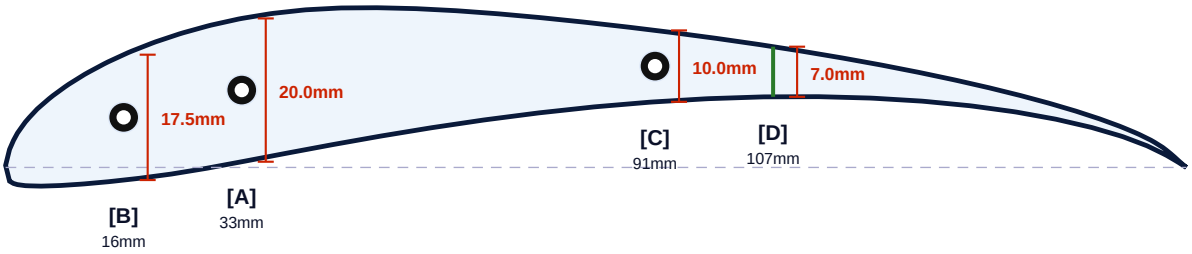

VG
17%c
[A]
33mm (20%c)
t=20.0mm
[B]
16mm (10%c)
t=17.5mm
[C]
91mm (55%c)
t=10.0mm
[D]
107mm (65%c)
D-BOX (LE – 65%c)
100gsm CF skin · 3 CF spars
camber line
RIB TAIL (65 – 100%c)
Ripstop nylon 40d
GURNEY 1.65mm
0.5chord flap
165 mm CHORD
[G] TE ROD
Ø0.3mm CF rod



S1223 COORDINATES AT SPAR STATIONS (UIUC · 165mm chord)

x %	5%	**10%**	15%	**20%**	25%	35%	45%	**55%**	60%	65%	75%	85%	100%
x mm	8.2	**16.5**	24.8	**33.0**	41.2	57.7	74.2	**90.8**	99.0	107.2	123.8	140.2	165.0
upper mm	+11.22	**+15.75**	+18.80	**+20.82**	+21.95	+22.17	+20.98	**+19.15**	+18.08	+16.87	+14.04	+10.43	+0.00
lower mm	-2.49	**-1.76**	-0.66	**+0.80**	+2.36	+5.26	+7.59	**+9.17**	+9.63	+9.87	+9.52	+7.88	+0.00
thick mm	13.71	**17.52**	19.46	**20.02**	19.60	16.91	13.39	**9.98**	8.44	7.01	4.51	2.56	0.00
camber mm	4.37	**7.00**	9.07	**10.81**	12.16	13.71	14.28	**14.16**	13.86	13.37	11.78	9.16	0.00

Bold/blue = spar stations · +y values = ABOVE chord line · Lower surface is positive from ~17%c onward (S1223 characteristic: deeply cambered under-camber)

AEROFOIL	
Profile	S1223 — UIUC Airfoil Database (m-selig.ae.illinois.edu/ads/coord/s1223.dat) Selig 1997
Chord	165 mm (locked Rev H)
Max thickness	12.1%c at 19.8%c → 20.0mm at 32.7mm from LE
Max camber	8.1%c at 49%c → 13.4mm at 80.9mm from LE
Lower surface note	POSITIVE (above chord line) from ~17%c to TE — deeply cambered under-camber (S1223 characteristic)
CL,max clean	2.20 (Re=2×10■, Selig 1997)
CL,max + VGs + Gurney	2.40

STRUCTURAL MEMBERS	
[A] Main spar	2.0mm OD / 1.0mm ID pultruded CF tube · 20%c = 33mm · thickness at station 20.0mm · ø2.2mm rib hole · thin CA
[B] Fwd spar	2.0mm OD / 1.0mm ID pultruded CF tube · 10%c = 17mm · thickness at station 17.5mm · ø2.2mm rib hole · thin CA
[C] Rear spar	2.0mm OD / 1.0mm ID pultruded CF tube · 55%c = 91mm · thickness at station 10.0mm · ø2.2mm rib hole · thin CA
[D] D-box rear wall	65%c = 107mm · full-depth LW-PLA element · 0.5mm wall · 3 perimeters · 20% infill · closes torsion box
[G] TE rod	0.3mm solid CF rod at trailing edge · ø0.5mm rib hole · E6000 hem bond · membrane anchor + TE stiffness + Gurney alignment

SKIN AND COVERING	
CF skin (D-box)	100gsm 2×2 twill CF woven · 1 ply upper + lower · LE to 65%c · West System epoxy · 0/90° to chord
Membrane (rib tail)	Ripstop nylon 40 denier · E6000 to rib tail edges · 65–100%c both surfaces · TE hemmed around [G] rod

RIB CONSTRUCTION	
Material	LW-PLA · gyroid 12% infill · 3 perimeters
Standard rib wall	1.0mm (Rev H — ribs are shape-limited; peak load 0.79N in sprint — 1.0mm adequate, saves 84g)
Root rib	3.0mm solid LW-PLA
Tip rib	2.0mm solid LW-PLA
Rib spacing	20mm (17%c) · membrane deflection at cruise: 0.03mm
Lightening hole	35×9mm oval between [B] and [A] spars

AERODYNAMIC FEATURES	
VGs	17%c upper surface · h=8.4mm · 18° toe · counter-rotating pairs · LW-PLA strip · E6000
Gurney flap	1.65mm (1%c) · integral LW-PLA · lower surface at TE · perpendicular to chord
Covert flaps	0.125mm PET · 50–62%c upper · 3 strips 15mm apart · E6000 hinge at 50%c · passive
Raked wingtip	25° LE sweep · 15%c (25mm) ext. · 1.5° washout · 5° dihedral

WING PAIRS	
W1 (front)	Semi-span 772mm · tip-to-tip 1544mm · area 0.255m²
W2	Semi-span 749mm · tip-to-tip 1498mm · area 0.247m²
W3	Semi-span 735mm · tip-to-tip 1470mm · area 0.243m²
W4 (rear)	Semi-span 712mm · tip-to-tip 1424mm · area 0.235m²
Total	Area 0.979m² · CL,max 2.40 · Vs=4.85m/s at 3.45kg MTOW · stagger 138mm